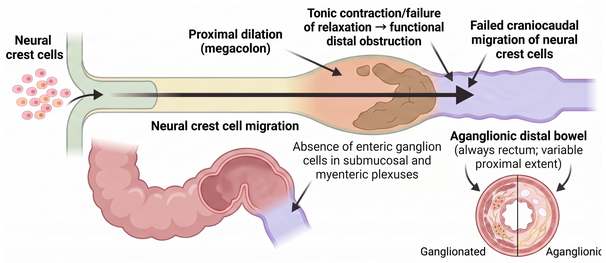
Targeted LC-MS/MS Measurement of the AChE/Calretinin Ratio in FFPE Colonic Tissue for Hirschsprung Disease Diagnosis

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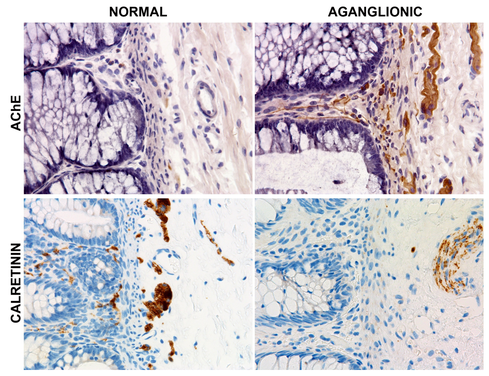
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BACKGROUND

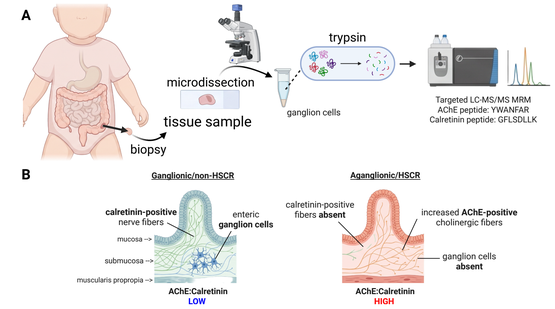
Hirschsprung disease (HSCR) is a congenital enteric nervous system disorder defined by **absence of ganglion cells in the distal bowel**, causing functional obstruction. Diagnosis depends on histopathologic evaluation of rectal biopsies, supported by **acetylcholinesterase (AChE)** histochemistry and **calretinin** immunohistochemistry, which suffer practical limitations. AChE histochemistry depends on **preserved enzyme activity** and typically requires **fresh or frozen tissue**, while calretinin immunohistochemistry is performed on **FFPE** tissue but remains dependent on staining conditions and **subjective interpretation**. Small biopsies and tissue near transition zones can be especially challenging.



Pathophysiology of Hirschsprung disease: During embryonic development, nerve-forming cells fail to migrate to the end of the large intestine. Affected bowel lacks enteric ganglion cells and calretinin-positive nerve fibers, causing functional obstruction with upstream dilation. [image: BioRender]



Hirschsprung disease pathology: AChE and calretinin show reciprocal patterns, with high AChE and low calretinin in aganglionic HSCR tissue. [image: Dr. Raj Kapur]



Assay workflow and biological rationale: (A) FFPE rectal biopsy tissue is microdissected, trypsin-digested, and analyzed by targeted LC-MS/MS for AChE + calretinin peptides. **(B)** Ganglionic tissue contains calretinin-positive nerve fibers and ganglion cells (lower AChE:calretinin), vs. aganglionic HSCR tissue lacks ganglion cells and shows increased AChE-positive cholinergic fibers, producing a higher ratio. [image: BioRender]

METHODS

- Developed **targeted LC-MS/MS assay** to measure **AChE:calretinin** in **FFPE colon tissue.**
- Tested whether **ACHE:calretinin ratio** distinguishes HSCR from non-HSCR biopsies.
- AChE and calretinin **surrogate peptides** selected via:
  1. *In-silico* prediction in Skyline
  2. Nano-flow LC-PRM screening (Thermo QE+)
  3. Clinical-flow LC-MRM test (Xevo TQ Absolute)
  4. Selected final peptides:
     - **AChE Peptide: YWANFAR**
     - **Calretinin Peptide: GFLSDLLK**
- **Assay workflow:**
  1. FFPE biopsy tissue
  2. mucosa/submucosa microdissection
  3. protein extraction
  4. trypsin digestion
  5. LC-MS/MS MRM of digest
  6. AChE/calretinin ratio-based classification
- **Method Validation:**
  - Linearity - Colon/muscle digest dilution
  - Imprecision - 3-day replicate testing
  - Carryover - High spike -> blank
  - Stability - 4 C hold + freeze-thaw
  - Accuracy - LC-MS/MS vs. IHC/pathology

CONCLUSION

An FFPE-compatible LC-MS/MS assay quantified the AChE/calretinin ratio in rectal biopsy tissue and classified adequate blinded specimens with 98.2% concordance.

RESULTS: PEPTIDE SELECTION

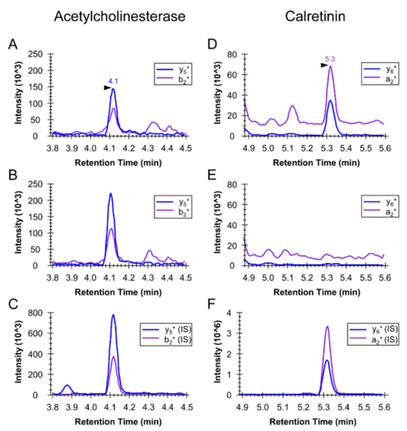
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Acetylcholesterase (P22303, AChE)

   1  MRPPQCLLHT PSLASPLLLL LLWLLGGGVG AEGREDAELL VTVRGGRLRG
  50  IRLKTPGGPV SAFLGIPFAE PPMGPRRFLP PEPKQPWSGV VDATTFQSVC
 100  YQYVDTLYPG FEGTEMWNPN RELSEDCLYL NVWTPYPRPT SPTPVLVWIY
 150  GGGFYSGASS LDVYDGRFLV QAERTVLVSM NYRVGAFGFL ALPGSREAPG
 200  NVGLLDQRLA LQWVQENVAA FGGDPTSVTL FGESAGAASV GMHLLSPPSR
 250  GLFHRAVLQS GAPNGPWATV GMGEARRRAT QLAHLVGCPP GGTGGNDTEL
 300  VACLRTRPAQ VLVNHEWHVL PQESVFRFSF VPVVDGDFLS DTPEALINAG
 350  DFHGLQVLVG VVKDEGSYFL VYGAPGFSKD NESLISRAEF LAGVRVGVPQ
 400  VSDLAAEAVV LHYTDWLHPE DPARLREALS DVVGDHNVVC PVAQLAGRLA
 450  AQGARVYAYV FEHRASTLSW PLWMGVPHGY EIEFIFGIPL DPSRNYTAEE
 500  KIFAQRLMRY WANFARTGDP NEPRDPKAPQ WPPYTAGAQQ YVSLDLRPLE
 550  VRRGLRAQAC AFWNRFLPKL LSATDTLDEA ERQWKAEFHR WSSYMVHWKN
 600  QFDHYSKQDR CSDL

Calritinin (P22676)

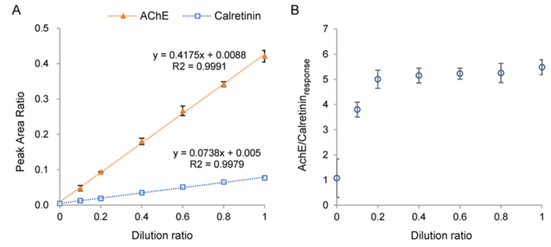
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  50  ARKGSGMMSK SDNFGEKMKE FMQKYDKNSD GKIEMAELAQ ILPTEENFLL
 100  CFRQHVGSSA EFMEAWRKYD TDRSGYIEAN ELKGFLSDLL KKANRPYDEP
 150  KLQEYTQTIL RMFDLNGDGK LGLSEMSRLL PVQENFLLKF QGMKLTSEEF
 200  NAIFTFYDKD RSGYIDEHEL DALLKDLYEK NKKEMNIQQL TNYRKSVMSL
```

Amino acid sequence: acetylcholinesterase and calretinin. Peptides selected for protein quantification in bold (**YWANFAR** and **GFLSDLLLK**).



Representative chromatograms. (**A,D**) Patient without HSCR disease. (**B,E**) Patient with HSCR disease. (**C,F**) Internal standards in a patient without SCR disease. Blue and purple traces represent quantitative and qualitative ion transitions, respectively.

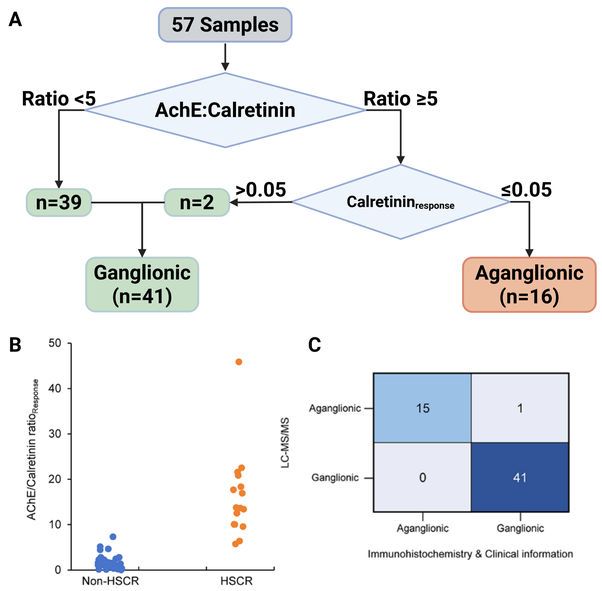
RESULTS: METHOD VALIDATION (SELECT STUDIES)



Linearity study: non-HSCR colon digest was mixed with low-analyte muscle digest at 0%, 10%, 25%, 40%, 60%, 75%, and 100% colon. AChE and calretinin responses were linear, while the AChE/calretinin ratio remained stable.

		Day 1	Day 2	Day 3	Mean[d]
AChE/calretinin ratio$_{Area}$ (without internal standard)[b]	**Sample 1**				
	Slide 1	5.6%	9.2%	5.4%	
	Slide 2	11.1%	15.1%	5.6%	
	Slide 3	14.8%	13.2%	4.0%	9.5%
	Slide 4	15.5%	7.1%	3.9%	
	Slide 5	16.0%	14.6%	1.3%	
	Sample 2				
	Slide 1	2.3%	19.6%	5.7%	
	Slide 2	7.8%	29.5%	6.4%	
	Slide 3	3.5%	14.3%	2.8%	9.3%
	Slide 4	2.9%	17.4%	8.0%	
	Slide 5	5.3%	8.3%	5.5%	
AChE/calretinin ratio$_{Response}$ (with internal standard)[c]	**Sample 1**				
	Slide 1	2.8%	5.4%	7.0%	
	Slide 2	4.1%	6.6%	11.4%	
	Slide 3	1.2%	10.8%	7.7%	5.4%
	Slide 4	0.9%	8.4%	6.3%	
	Slide 5	3.5%	1.1%	4.4%	
	Sample 2				
	Slide 1	2.0%	5.4%	12.3%	
	Slide 2	7.7%	21.5%	12.8%	
	Slide 3	2.4%	8.1%	9.0%	7.0%
	Slide 4	1.9%	2.4%	4.6%	
	Slide 5	7.2%	4.1%	3.3%	

Precision study: Assessed by triplicate daily injections of processed samples. Raw peak-area and internal-standard- normalized AChE/calretinin ratios were compared, with mean %CV calculated across 15 samples.



Clinical evaluation: (**A**)The AChE/calretinin ratio$_{Response}$ and calretinin response (peak area ratio) were evaluated in blinded biopsies and used to categorize specimens as ganglionic (non-HSCR) and aganglionic (HSCR). (**B**) Observed AChE/calretinin ratio$_{Response}$ in blinded specimens. (**C**) Concordance diagram for diagnosis by LC-MS/MS vs. diagnosis by immunohistochemistry and clinical information.

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Conflict of Interest: The University of Washington (ANH) receives grant and equipment support from Waters, Inc., a mass spectrometry company.

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